Wael Halabi

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EDUCATION

Bachelor of Applied Science and Engineering (B.A.Sc) in Civil Engineering September 2022 to April 2027
University of Toronto, Toronto, ON CGPA: 3.50

- **Relevant Courses**: CIV331 Transport I: Introduction to Transportation Systems, CIV332 Transport II: Performance, CIV531 Transport III: Planning, CIV516 Public Transportation Operations & Planning, CSC311 Machine Learning
- **Awards**: Dean's List (3x), Manuel A. Fine Scholarship, WSP Scholarship in Building Engineering, Victor Xin Scholarship in Esports, Honors, Winner of the Translational AIxBio Hackathon (ShiftED), 10th of 328 participants at the NSRI Summer Research Hackathon

SKILLS

Languages and Libraries: **Python** (Pandas, NumPy, SciPy, scikit-learn, Matplotlib, psycopg2, SQLAlchemy), **SQL** (PostgreSQL), **R**
GIS, Design, and Modelling Software: EMME, MATSim, OpenPath, ArcGIS, QGIS, Autodesk Revit, AutoCAD, Rhino 3D, SketchUp
Tools and Platforms: Git, GitHub, DBeaver, SSH, AWS, LaTeX
Other Skills: Microsoft Suite (Excel, Word, PowerPoint, Flow, Power Automate), Project Management, Arabic Fluency

EXPERIENCE

City of Toronto September 2025 to September 2026
Transportation Services Data Science Intern Toronto, Canada

- **Assisted in developing the City's bicycle AADT estimation framework**, combining multi-source traffic counts, temporal normalization, seasonal adjustment, and confidence interval estimation to support transportation planning.
- **Built automated congestion monitoring workflows** supporting **FIFA World Cup** transportation planning, analyzing traffic volumes, travel times, and corridor performance to monitor network conditions and support operational decision-making.
- **Designed an analytics pipeline evaluating Automated Speed Enforcement (ASE) camera removals**; analyses informed City Council reporting and public communications discussed by the Mayor.
- Developed geospatial accessibility models in R and QGIS, along with machine learning models for **traffic congestion forecasting** and **collision severity prediction**.
- Worked across all four groups of the Data & Analytics Unit (data collection, data operations, data analytics, and internal app development), using SQL, PostgreSQL, and Python on sensor data systems.

Saudi Icon May 2024 to July 2024
Civil Engineering Intern Riyadh, Saudi Arabia

- Supported site engineering for a **9-floor building** and external infrastructure project, collaborating with foremen and engineers to uphold safety and compliance standards.
- **Supervised concrete testing**, verified materials met local engineering standards, and **reviewed blueprints** to identify construction risks.
- Managed communication between on-site teams and management to **resolve workflow bottlenecks** and ensure timely project delivery.

PROJECTS

Predicting Streetcar Delays in Toronto Using Historical Data and Machine Learning August 2025

- **Analyzed 10+ years of TTC streetcar transit data** to identify delay patterns across multiple routes.
- Built and validated Linear Regression and Random Forest models in Python (Pandas, scikit-learn) to **predict delay durations**, with Random Forest outperforming on complex, non-linear relationships across evaluated routes.
- Produced a route-level performance report surfacing actionable insights for data-driven scheduling and service optimization.

Analysis of the 510 Streetcar Route: Patterns and Operational Dynamics Jan – May 2024

- Conducted comprehensive analysis of Toronto 510 Streetcar operations over 3 months, collecting **150+ multi-point data entries** to inform operational improvements; published findings on Medium.
- Filtered and processed data using Python and Pandas, producing visual reports on traffic flow and scheduling patterns.

EXTRACURRICULARS

Canadian Society for Civil Engineering (CSCE), University of Toronto Chapter November 2023 to Present
Co-President (April 2025 to May 2026) | VP Academia (November 2023 to April 2025)

- Led a six-sector executive team supporting over 300 engineering students through professional development, networking, and academic initiatives.
- Secured over **$4,000** in sponsorships and managed annual event budgets exceeding **$14,000**.
- Founded the External Relations Team and led more than 20 executive interviews to strengthen long-term organizational growth.